

Cutoff describes a phase transition: as we run the family of Markov chains on X_n , the total variation distance is almost equal to 1, and then suddenly it drops and approaches zero as n goes to infinity. The formal definition of cutoff is given below.

Definition 2. A family of Markov chains on X_n is said to have cutoff at time t_n with window $w_n = o(t_n)$ if and only if

$$\lim_{c \rightarrow \infty} \lim_{n \rightarrow \infty} d_x^{(n)}(t_n - cw_n) = 1 \text{ and } \lim_{c \rightarrow \infty} \lim_{n \rightarrow \infty} d_x^{(n)}(t_n + cw_n) = 0,$$

where $d_x^{(n)}(t)$ denotes the total variation distance of the n -th Markov chain after t steps starting at x .

Given a Markov chain exhibiting cutoff, one can ask for more precise control on the exact distance from stationarity. This is known as the limit profile with respect to the sequences (t_n) and (w_n) , defined as:

$$\Phi_x(c) := \lim_{n \rightarrow \infty} d_x^{(n)}(t_n + cw_n), \text{ for all } c \in \mathbb{R}, \quad (1)$$

if this limit exists.

The limit profile is known for only a few Markov chains, such as the riffle shuffle [2], the asymmetric exclusion process on the segment [4], the simple exclusion process on the cycle [14], and the simple random walk on Ramanujan graphs [16], etc. Teyssier [25] determined the limit profile for random transpositions. Using representation theory of the symmetric group S_n , he used Fourier transform arguments for studying limit profiles that work for random walks on groups using a generating set that is a conjugacy class. The same limit profile is proved to be true for the k -cycle card shuffle [18], under the assumption that $k = o(n)$ and for star transposition [17]. Olesker-Taylor and Schmid also studied the limit profile for the Bernoulli-Laplace chain [21]. In Section 6 we give a more extensive presentation of the above results.

Furthermore, if P is aperiodic, irreducible and reversible on X , then the spectrum of P satisfies

$$-1 < \beta_{|X|} \leq \dots \leq \beta_2 < \beta_1 = 1,$$

and there is an orthonormal eigenbasis $\{f_j : X_n \rightarrow \mathbb{R}\}$. The main assumption on P is that there is a continuous (or bounded) function g on $\mathbb{R}$ such that

$$\limsup_{n \rightarrow \infty} w_n^2 \sum_{i=2}^{|X_n|} f_i(x)^2 (1 - \beta_i)^2 e^{-2(t_n + cw_n)(1 - \beta_i)} \leq g(c), \quad (2)$$

for all $c \in \mathbb{R}$.

Theorem 3. Let P_n be the transition matrix of a reversible Markov chain on X_n that exhibits cutoff at t_n with window w_n and satisfies (2). Assume that Φ_x , the total variation limit profile with respect to the sequences (t_n) and (w_n) at x , exists. Then Φ_x is continuous for every $x \in X$.